DESSERTOPIA

SQL QUERY IMPLEMENTATION FOR E-COMMERCE DATABASE REPORT

Introduction:

This report presents the SQL queries implemented in the Dessertopia E-Commerce Database. The database stores customer, product, order, payment, inventory, and review information. SQL is used to retrieve, filter, sort, and analyse data for daily operations and basic business reporting.

Displaying Product IDs and Product Names in Ascending Order :

```
SQL> SELECT PRODUCT_ID, PRODUCT_N
  2  FROM PRODUCT
  3  ORDER BY PRODUCT_ID;

PRODUCT PRODUCT_NAME
------- --------------------
P001    CHOCOLATE CAKE
P002    VANILLA CUPCAKE
P003    CHOCO BROWNIE
P004    BUTTER COOKIES
P005    GLAZED DONUT
P006    BLUEBERRY CHEESECAKE
P007    RED VELVET ICE CREAM
```

Retrieving Product IDs from the PRODUCT Table :

```
SQL> SELECT PRODUCT_ID FROM PRODUCT;

PRODUCT
-------
P001
P002
P003
P004
P005
P006
P007

7 rows selected.
```

Describing the Structure of the CUSTOMER Table :

```
SQL> DESC CUSTOMER;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 CUSTOMER_ID                               NOT NULL VARCHAR2(10)
 FIRST_NAME                                NOT NULL VARCHAR2(30)
 EMAIL_ID                                  NOT NULL VARCHAR2(30)
 PHONE_NUMBER                              NOT NULL NUMBER(10)
 ADDRESS                                            VARCHAR2(100)
 CREATED_AT                                         DATE
 LAST_NAME                                 NOT NULL VARCHAR2(20)
```

Displaying Products with Available Stock Quantity :

```
SQL> SELECT * FROM PRODUCT WHERE STOCK_QUANTITY > 0;

PRODUCT PRODUCT_NAME         CATEGOR      PRICE STOCK_QUANTITY
------- -------------------- ------- ---------- --------------
P001    CHOCOLATE CAKE       CAT001         550             20
P002    VANILLA CUPCAKE      CAT003         120             30
P003    CHOCO BROWNIE        CAT003         180             25
P004    BUTTER COOKIES       CAT004         150             40
P005    GLAZED DONUT         CAT005         100             35
P006    BLUEBERRY CHEESECAKE CAT006         450             20
P007    RED VELVET ICE CREAM CAT007         200             25

7 rows selected.
```

Retrieving Product Names and Their Prices :

```
SQL> SELECT PRODUCT_NAME , PRICE FROM PRODUCT;

PRODUCT_NAME              PRICE
-------------------- ----------
CHOCOLATE CAKE              550
VANILLA CUPCAKE             120
CHOCO BROWNIE               180
BUTTER COOKIES              150
GLAZED DONUT                100
BLUEBERRY CHEESECAKE        450
RED VELVET ICE CREAM        200

7 rows selected.
```

Displaying All Records from the ORDERS Table :

```
SQL> SELECT * FROM ORDERS;

ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- ---------- --------- ------------ --------------------
O001       C001       01-OCT-26          550 Pending
O002       C002       02-OCT-26          240 Shipped
O003       C003       03-OCT-26          360 Delivered
O004       C004       04-OCT-26          300 Pending
O005       C005       05-OCT-26          500 Shipped
O006       C006       06-OCT-26          450 Delivered
O007       C007       07-OCT-26          400 Pending

7 rows selected.
```

Displaying All Payment Transaction Records:

```
SQL> SELECT * FROM PAYMENT ;

PAYMENT_ID ORDER_ID   PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
PAY001     O001       01-OCT-26 UPI                  Success
               550

PAY002     O002       02-OCT-26 Card                 Success
               240

PAY003     O003       03-OCT-26 Cash                 Failed
               360


PAYMENT_ID ORDER_ID   PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
PAY004     O004       04-OCT-26 Net Banking          Pending
               300

PAY005     O005       05-OCT-26 UPI                  Success
               500

PAY006     O006       06-OCT-26 Card                 Failed
               450


PAYMENT_ID ORDER_ID   PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
PAY007     O007       07-OCT-26 UPI                  Success
               400


7 rows selected.
```

Retrieving Products Priced Below ₹250 :

```
SQL> SELECT * FROM PRODUCT WHERE PRICE < 250 ;

PRODUCT PRODUCT_NAME         CATEGOR      PRICE STOCK_QUANTITY
------- -------------------- ------- ---------- --------------
P002    VANILLA CUPCAKE      CAT003         120             30
P003    CHOCO BROWNIE        CAT003         180             25
P004    BUTTER COOKIES       CAT004         150             40
P005    GLAZED DONUT         CAT005         100             35
P007    RED VELVET ICE CREAM CAT007         200             25
```

Displaying Inventory Records with Available Stock Status :

```
SQL> SELECT *
  2  FROM INVENTORY
  3  WHERE STOCK_STATUS = 'Available';

INVENTO PRODUCT SELLER_ STOCK_QUANTITY STOCK_STATU LAST_UPDA
------- ------- ------- -------------- ----------- ---------
I001    P001    S001                20 Available   01-OCT-26
I002    P002    S002                30 Available   02-OCT-26
I003    P003    S003                25 Available   03-OCT-26
I005    P005    S005                35 Available   05-OCT-26
```

Sorting Products by Price in Ascending Order :

```
SQL> SELECT PRODUCT_NAME,PRICE FROM PRODUCT ORDER BY PRICE ASC;

PRODUCT_NAME              PRICE
-------------------- ----------
GLAZED DONUT                100
VANILLA CUPCAKE             120
BUTTER COOKIES              150
CHOCO BROWNIE               180
RED VELVET ICE CREAM        200
BLUEBERRY CHEESECAKE        450
CHOCOLATE CAKE              550

7 rows selected.
```

Displaying Customer Records in Alphabetical Order :

```
SQL> SELECT * FROM CUSTOMER ORDER BY FIRST_NAME ASC;

CUSTOMER_I FIRST_NAME                    EMAIL_ID
---------- ------------------------------ ------------------------------
PHONE_NUMBER
------------
ADDRESS
--------------------------------------------------------------------------------
CREATED_A LAST_NAME
--------- --------------------
C003       Ananya                        ananya@gmail.com
  9876543203
Coimbatore
03-OCT-26 Krishnan


CUSTOMER_I FIRST_NAME                    EMAIL_ID
---------- ------------------------------ ------------------------------
PHONE_NUMBER
------------
ADDRESS
--------------------------------------------------------------------------------
CREATED_A LAST_NAME
--------- --------------------
C006       Divya                         divya@gmail.com
  9876543206
Chennai
06-OCT-26 Raj
```

Displaying Customer Records Sorted by First Name – Continued Output :

```
CUSTOMER_I FIRST_NAME                    EMAIL_ID
---------- ------------------------------ ------------------------------
PHONE_NUMBER
------------
ADDRESS
--------------------------------------------------------------------------------
CREATED_A LAST_NAME
--------- --------------------
C005       Harini                        harini@gmail.com
  9876543205
Bengaluru
05-OCT-26 Kumar


CUSTOMER_I FIRST_NAME                    EMAIL_ID
---------- ------------------------------ ------------------------------
PHONE_NUMBER
------------
ADDRESS
--------------------------------------------------------------------------------
CREATED_A LAST_NAME
--------- --------------------
C001       Kavya                         kavya@gmail.com
  9876543201
Chennai
01-OCT-26 Dharshini


CUSTOMER_I FIRST_NAME                    EMAIL_ID
---------- ------------------------------ ------------------------------
PHONE_NUMBER
------------
ADDRESS
--------------------------------------------------------------------------------
CREATED_A LAST_NAME
--------- --------------------
C004       Priya                         priya@gmail.com
  9876543204
Madurai
04-OCT-26 Sharma
```

```
CUSTOMER_I FIRST_NAME                    EMAIL_ID
---------- ----------------------------- ------------------------------
PHONE_NUMBER
------------
ADDRESS
--------------------------------------------------------------------------------
CREATED_A LAST_NAME
--------- --------------------
C002       Roshini                       roshini@gmail.com
  9876543202
Chennai
02-OCT-26 Priya


CUSTOMER_I FIRST_NAME                    EMAIL_ID
---------- ----------------------------- ------------------------------
PHONE_NUMBER
------------
ADDRESS
--------------------------------------------------------------------------------
CREATED_A LAST_NAME
--------- --------------------
C007       Sneha                         sneha@gmail.com
  9876543207
Coimbatore
07-OCT-26 Mohan


7 rows selected.
```

Sorting Orders by Order Date in Descending Order :

```
SQL> SELECT * FROM ORDERS ORDER BY ORDER_DATE DESC;

ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- ---------- --------- ------------ --------------------
O007       C007       07-OCT-26          400 Pending
O006       C006       06-OCT-26          450 Delivered
O005       C005       05-OCT-26          500 Shipped
O004       C004       04-OCT-26          300 Pending
O003       C003       03-OCT-26          360 Delivered
O002       C002       02-OCT-26          240 Shipped
O001       C001       01-OCT-26          550 Pending

7 rows selected.
```

SQL query filtering products priced between 100 and 500 inclusive :

```
SQL> SELECT * FROM PRODUCT WHERE PRICE BETWEEN 100 AND 500;

PRODUCT PRODUCT_NAME         CATEGOR      PRICE STOCK_QUANTITY
------- -------------------- ------- ---------- --------------
P002    VANILLA CUPCAKE      CAT003         120             30
P003    CHOCO BROWNIE        CAT003         180             25
P004    BUTTER COOKIES       CAT004         150             40
P005    GLAZED DONUT         CAT005         100             35
P006    BLUEBERRY CHEESECAKE CAT006         450             20
P007    RED VELVET ICE CREAM CAT007         200             25

6 rows selected.
```

Distinct product category identifiers retrieved from the PRODUCT table.

```
SQL> SELECT DISTINCT CATEGORY_ID FROM PRODUCT;

CATEGOR
-------
CAT003
CAT007
CAT006
CAT004
CAT001
CAT005

6 rows selected.
```

Fetching distinct product category names by joining the CATEGORY and PRODUCT tables :

```
SQL> SELECT DISTINCT C.CATEGORY_NAME
  2  FROM CATEGORY C
  3  JOIN PRODUCT P
  4  ON C.CATEGORY_ID = P.CATEGORY_ID;

CATEGORY_NAME
------------------------------
CAKES
DONUTS
COOKIES
CHEESECAKES
ICE CREAMS
BROWNIES

6 rows selected.
```

Unique payment methods retrieved from transaction records :

```
SQL> SELECT DISTINCT PAYMENT_MODE FROM PAYMENT;

PAYMENT_MODE
--------------------
Net Banking
Card
Cash
UPI
```